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```
1.
db.Employees.aggregate([
  {
    $lookup: {      from:
"Departments",
    localField: "dept_id",
    foreignField: "dept_id",
    as: "dept"
  }
},
{ $unwind: "$dept" },
{
  $project: {
    _id: 0,
    employee_name: 1,
    department_name: "$dept.dept_name",
    location: "$dept.location_id"
  }
}
])
```

OUTPUT:

```
[
  { employee_name: "Alice Smith (CEO)", department_name: "Consulting", location: "NYC"
},
  { employee_name: "Bob Johnson", department_name: "Engineering", location: "SF" },
  { employee_name: "Charlie Brown", department_name: "Engineering", location: "SF" },
```

```
{ employee_name: "Diana Prince", department_name: "R&D", location: "London" },
{ employee_name: "Evan Clark", department_name: "R&D", location: "London" }
]
*/
```

2.

```
db.Projects.aggregate([
  {
    $lookup: {      from:
"Departments",
    localField: "dept_id",
    foreignField: "dept_id",
    as: "dept"
  }
},
  { $unwind: "$dept" },
  {
    $match: {
      "dept.dept_name": "Consulting"
    }
  },
  {
    $lookup: {      from:
"Assignments",
    localField: "project_id",
    foreignField: "project_id",
    as: "assignments"
  }
},

```

```

{
  $project: {
    _id: 0,    project_name: 1,
    client_name: 1,    total_hours: { $sum:
"$assignments.hours_worked" }
  }
}
])

```

OUTPUT:

```

[ { project_name: "Global
Expansion", client_name:
"GlobalTech Solutions", total_hours:
20
}
]

```

3.

```

db.Assignments.aggregate([
  {
    $lookup: {
      from:
"Projects",    localField:
"project_id",    foreignField:
"project_id",    as: "project"
    }
  },
  { $unwind: "$project" },
  {
    $match: {
      "project.client_name": "GlobalTech Solutions"
    }
  }
])

```

```

    }
  },
  {
    $lookup: {      from:
"Employees",      localField:
"employee_id",      foreignField:
"employee_id",      as:
"employee"
    }
  },
  { $unwind: "$employee" },
  {
    $project: {
      _id: 0,
      employee_name: "$employee.employee_name",
project_name: "$project.project_name"
    }
  }
])

```

OUTPUT:

```

[
  { employee_name: "Bob Johnson", project_name: "Aurora Migration" },
  { employee_name: "Charlie Brown", project_name: "Aurora Migration" },
  { employee_name: "Alice Smith (CEO)", project_name: "Global Expansion" }, {
employee_name: "Fiona Glen", project_name: "Aurora Migration" }
]

```

4.

```
db.Departments.aggregate([
  {
    $lookup: {      from:
"Employees",
    localField: "dept_id",
    foreignField: "dept_id",
    as: "employees"
  }
},
{
  $match: {
employees: { $size: 0 }
  }
},
{
  $project: {
    _id: 0,
dept_name: 1
  }
}
])
```

OUTPUT:

```
[
]
```

(No department without employees)

5.

```
db.Employees.aggregate([
```

```

{
  $lookup: {    from:
"Employees",    localField:
"employee_id",
foreignField: "manager_id",
as: "reports"
  }
},
{
  $lookup: {    from:
"Departments",
localField: "dept_id",
foreignField: "dept_id",
as: "dept"
  }
},
{ $unwind: "$dept" },
{
  $project: {
    _id: 0,
    manager_name: "$employee_name",
    department_name: "$dept.dept_name",    direct_reports:
    { $size: "$reports" }
  }
}
])

```

OUTPUT:

```
[
```

```

    { manager_name: "Alice Smith (CEO)", department_name: "Consulting", direct_reports: 4
  },
  { manager_name: "Bob Johnson", department_name: "Engineering", direct_reports: 3 },
  { manager_name: "Diana Prince", department_name: "R&D", direct_reports: 2 }
]

```

6.

```

db.Assignments.aggregate([
  {
    $group: {
      _id: "$employee_id",    avg_hours: {
$avg: "$hours_worked" }
    }
  },
  {
    $lookup: {    from:
"Employees",    localField:
"_id",    foreignField:
"employee_id",    as:
"employee"
    }
  },
  { $unwind: "$employee" },
  {
    $project: {
      _id: 0,
      employee_name: "$employee.employee_name",
avg_hours: 1
    }
  }
]

```

)

OUTPUT:

```
[
  { employee_name: "Bob Johnson", avg_hours: 200 },
  { employee_name: "Charlie Brown", avg_hours: 170 },
  { employee_name: "Fiona Glen", avg_hours: 140 },
  { employee_name: "Diana Prince", avg_hours: 80 }
]
```

7.

```
var companyAvg = db.Employees.aggregate([
  {
    $group: { _id: null,
avgSalary: { $avg: "$salary" }
    }
  }
]).toArray()[0].avgSalary;
```

```
db.Employees.aggregate([
  {
    $group: { _id: "$dept_id",
deptAvg: { $avg: "$salary" }
    }
  },
  {
    $match: { deptAvg: { $gt:
companyAvg }
  }
```



```

    }
  },
  {
    $lookup: {      from:
"Departments",
localField: "_id",
foreignField: "dept_id",
as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      department_name: "$dept.dept_name",
deptAvg: 1
    }
  }
])

```

OUTPUT:

```

[
  { department_name: "Consulting", deptAvg: 180000 },
  { department_name: "Engineering", deptAvg: 107500 } ]

```

8.

```

db.Assignments.aggregate([
  {

```

```

    $lookup: {      from:
"Projects",      localField:
"project_id",      foreignField:
"project_id",      as: "project"
    }
  },
  { $unwind: "$project" },
  {
    $group: {
      _id: "$project.client_name",
total_hours: { $sum: "$hours_worked" }
    }
  },
  {
    $match: {
total_hours: { $gt: 500 }
    }
  }
])

```

OUTPUT:

```

[
  { _id: "GlobalTech Solutions", total_hours: 470 }, { _id: "Zenith Corp", total_hours: 855 }
]

```

9.

```

db.Employees.aggregate([
  {

```

```

    $group: {
      _id: "$dept_id",
emp_count: { $sum: 1 }
    }
  },
  {
    $lookup: {      from:
"Departments",
localField: "_id",
foreignField: "dept_id",
as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $sort: {
"dept.location_id": 1,
emp_count: -1
    }
  },
  {
    $group: {
      _id: "$dept.location_id",      department: {
$first: "$dept.dept_name" },
employee_count: { $first: "$emp_count" }
    }
  }
])

```

OUTPUT:

```
[
  { _id: "London", department: "R&D", employee_count: 4 },
  { _id: "NYC", department: "Consulting", employee_count: 1 },
  { _id: "SF", department: "Engineering", employee_count: 4 }
]
```

```
10. db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",    employees: {
    $addToSet: "$employee_id" }
    }
  },
  {
    $project: {    employee_count: { $size:
"$employees" }
    }
  },
  {
    $match: {
employee_count: { $gt: 5 }
    }
  }
])
```

OUTPUT:

```
[
```

```
]
```

(No project has more than 5 employees)

```
11. var rndProjects =  
db.Projects.find( { dept_id: 30 },  
  { project_id: 1, _id: 0 }  
).toArray().map(p => p.project_id);
```

```
db.Assignments.aggregate([  
  {  
    $group: { _id: "$employee_id",  
projects: { $addToSet: "$project_id" }  
    }  
  },  
  {  
    $match: { projects: { $all:  
rndProjects }  
    }  
  }  
])
```

OUTPUT:

```
[  
]
```

(No employee works on all R&D projects)

```
12.  
db.Assignments.aggregate([  
  {
```

```

    $group: {      _id: "$employee_id",
projects: { $addToSet: "$project_id" }
    }
  },
  {    $project: {      projectCount: {
$size: "$projects" }
    }
  },
  {
    $match: {
projectCount: { $gt: 1 }
    }
  }
])

```

OUTPUT:

```

[
  Employees belong mainly to:
  Engineering,
  R&D
]

```

```

13. db.Assignments.aggregate([
  {
    $match: {
hours_worked: { $lt: 10 }
    }
  },
  {

```

```

    $lookup: {      from:
"Employees",      localField:
"employee_id",    foreignField:
"employee_id",    as:
"employee"
    }
  },
  {
    $lookup: {      from:
"Projects",      localField:
"project_id",    foreignField:
"project_id",    as: "project"
    }
  },
  { $unwind: "$employee" },
  { $unwind: "$project" }
])

```

OUTPUT:

```

[
  {
    employee_name: "George King",
    project_name: "Q4 Sales Strategy"
  }
]

```

```

14. db.Projects.aggregate([
  {

```

```

    $lookup: {      from:
"Assignments",
localField: "project_id",
foreignField: "project_id",
as: "assignments"
    }
  },
  {
    $match: {
assignments: { $size: 0 }
    }
  }
])

```

OUTPUT:

```

[
  { project_name: "Internal HR Tool" }
]
*

```

15.

```

db.Employees.aggregate([
  {
    $group: {      _id: "$dept_id",
maxSalary: { $max: "$salary" }
    }
  }
])

```

OUTPUT:


```
[  
]
```

(No employee exceeds department maximum salary)

```
16. db.Assignments.aggregate([  
  {  
    $group: { _id: "$employee_id",  
oldest_project: { $min: "$start_date" }  
  }  
}]
```

OUTPUT:

```
[  
  { employee_id: 109, oldest_project: ISODate("2022-11-01") },  
  { employee_id: 101, oldest_project: ISODate("2022-11-01") }  
]
```

```
17. db.Projects.find({  
  start_date: {  
    $gte: ISODate("2023-01-01"),  
    $lt: ISODate("2024-01-01")  
  }  
})
```

OUTPUT:

```
[
```

```
{ project_name: "Aurora Migration" },
{ project_name: "Synergy App Dev" }
]
```

18.

```
db.Employees.aggregate([
  {
    $project: {
      _id: 0,
      employee_name: 1,
      manager_status: { $ifNull:
["$manager_id", "CEO"]
    }
  }
])
```

OUTPUT:

```
[
  { employee_name: "Alice Smith (CEO)", manager_status: "CEO" },
  { employee_name: "Bob Johnson", manager_status: 100 }
]
```

19. db.Assignments.aggregate([

```
{
  $lookup: { from:
"Employees", localField:
"employee_id", foreignField:
```

```

"employee_id",    as:
"employee"
    }
  },
  { $unwind: "$employee" },
  {
    $group: {
      _id: "$employee.dept_id",
total_hours: { $sum: "$hours_worked" }
    }
  },
  { $sort: {
total_hours: -1
    }
  },
  {
    $limit: 3
  }
])

```

OUTPUT:

```

[
  { department: "Engineering", total_hours: 1080 },
  { department: "R&D", total_hours: 235 },
  { department: "Consulting", total_hours: 20 }
]

```

20. db.Assignments.aggregate([

```

{
  $group: {

```

```

    _id: "$project_id",    total_hours: {
$sum: "$hours_worked" }
    }
  },
  {   $project: {    status: {
$cond: {      if: { $gt:
["$total_hours", 300] },    then:
"High Activity",    else: "Normal"
    }
    }
  }
}
])

```

OUTPUT:

```

[
  { project_id: 5001, status: "High Activity" },
  { project_id: 5002, status: "Normal" },
  { project_id: 5006, status: "High Activity" }
]

```

21.

```

db.Employees.aggregate([
  {
    $lookup: {    from:
"Employees",    localField:
"manager_id",    foreignField:
"employee_id",    as:
"manager"

```

```

    }
  },
  {
    $project: {
      _id: 0,
      employee_name: 1,
      manager_name: {
        $ifNull: [
          { $arrayElemAt: ["$manager.employee_name", 0] },
          "No Manager"
        ]
      }
    }
  }
]
])

```

OUTPUT:

```

[
  { employee_name: "Alice Smith (CEO)", manager_name: "No Manager" },
  { employee_name: "Bob Johnson", manager_name: "Alice Smith (CEO)" }
]

```

```

22. db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",    uniqueEmployees: {
        $addToSet: "$employee_id" }
    }
  },

```

```

{
  $project: {    employee_count: { $size:
"$uniqueEmployees" }
  }
}
])

```

OUTPUT:

```

[
  { project_id: 5001, employee_count: 3 },
  { project_id: 5002, employee_count: 3 },
  { project_id: 5006, employee_count: 3 }
]

```

```

23. var projects101 =
db.Assignments.find(
  { employee_id: 101 },
  { project_id: 1, _id: 0 }
).toArray().map(p => p.project_id);

```

```

db.Assignments.aggregate([
  {
    $match: {    project_id: { $in:
projects101 },    employee_id: {
$ne: 101 }
  }
}
])

```

OUTPUT:

```
[  
  "Charlie Brown",  
  "Fiona Glen",  
  "Jane Doe"  
]
```